

EE115 Analog Circuits

Ideal Op Amp

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Outline

- Operational Amplifiers
 - Ideal Op Amp
 - Inverting Configuration
 - Noninverting Configuration
 - Difference Amplifiers
- Reading: SEDTRA/SMITH book pages 58-82

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Ideal Op Amp

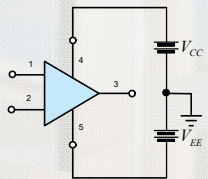


Fig. Op amp shown connected to dc power supplies.

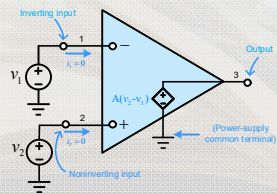


Fig. Equivalent circuit of the ideal op amp.

- Infinite input impedance
 - No current goes in, virtual open
- Zero output impedance
- Infinite open-loop gain, A
 - How is this possible?
 - Use feedback to define circuit gain
 - $v_1 = v_2$, virtual short
- Infinite bandwidth
- Infinite common-mode rejection

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Inverting Amplifier

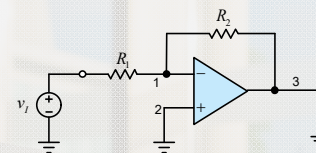
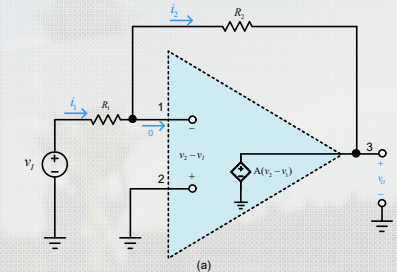
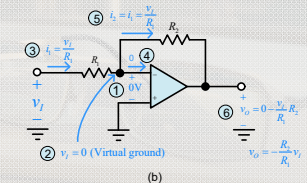


Fig. Inverting closed-loop configuration.

$$\frac{v_o}{v_I} = -\frac{R_2}{R_1}$$



(a)



(b)

Fig. Analysis of the inverting configuration.

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Inverting Amplifier with Finite Open-Loop Gain



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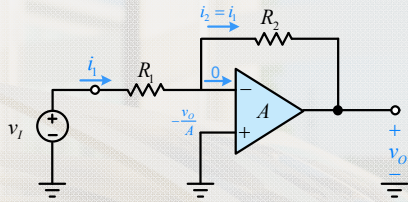


Figure 2.7 Analysis of the inverting configuration taking into account the finite open-loop gain of the op amp.

$$i_1 = \frac{v_I - (-v_O/A)}{R_1} = \frac{v_I + v_O/A}{R_1}$$

$$v_O = -\frac{v_O}{A} - i_1 R_2 = -\frac{v_O}{A} - \left(\frac{v_I + \frac{v_O}{A}}{R_1} \right) R_2$$

$$G \equiv \frac{v_O}{v_I} = \frac{-R_2/R_1}{1 + (1 + R_2/R_1)/A}$$

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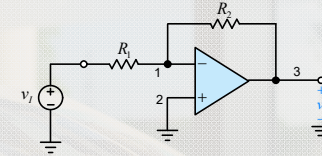
Inverting Amplifier: Input and Output Resistances



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■ Since $v_1 = 0$

➤ The input resistance is simply $R_i = R_1$



■ To find output resistance

➤ 1. Short-circuit the source

➤ 2. Apply a test source at the output, and find the impedance

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Application of Inverting AMP: Weighted Summer



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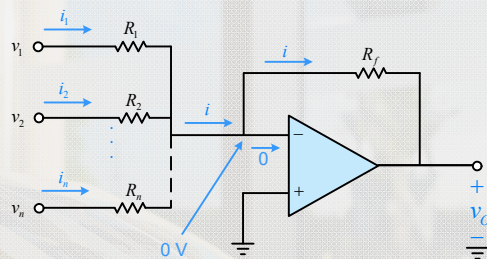


Fig. A weighted summer.

$$v_O = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \dots + \frac{R_f}{R_n} v_n \right)$$

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Non-Inverting Amplifier



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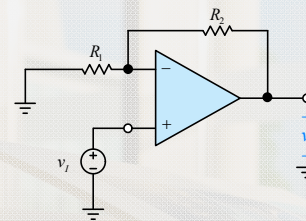


Fig. The noninverting configuration.

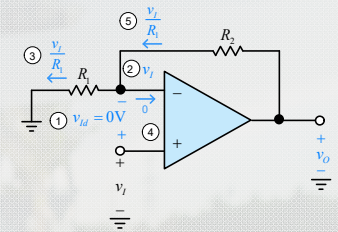


Fig. Analysis of the noninverting circuit.

$$\frac{v_O}{v_I} = 1 + \frac{R_2}{R_1}$$

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Non-Inverting AMP: Effect of Finite Open-Loop Gain

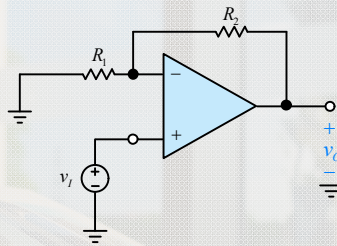


Fig. Noninverting configuration.

$$G \equiv \frac{v_O}{v_I} = \frac{1 + R_2/R_1}{1 + \frac{(1 + R_2/R_1)}{A}}$$

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Non-Inverting AMP Input and Output R



- For ideal Op-Amp
- No current flows into the amplifier
- $R_i = \infty$
- To find output resistance
 - Replace v_I with short circuit
- $R_o = 0$

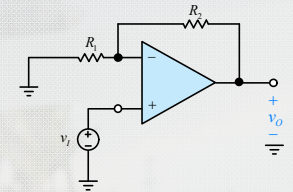


Fig. Noninverting configuration.

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Voltage Follower: Unity Gain Amplifier



- **Unity** voltage gain
 - But large power gain

Purpose:

- Provide **impedance transformation**:
 - A buffer stage that present infinite input impedance to the source, and zero output impedance to the load
- Power amplifier
 - Boost drive current

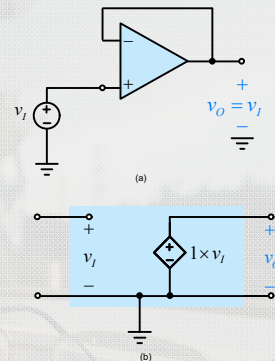


Fig. (a) The unity-gain buffer or follower amplifier. (b) Its equivalent circuit model.

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Differential Amplifier



- A differential amplifier is one that **responds to the difference** between the two signals applied at its input and ideally **rejects** signals that are **common** to the two inputs.

$$v_O = A_d v_{Id} + A_{cm} v_{Icm}$$

- Common-mode rejection ratio (CMRR)
- $CMRR = 20 \log(|A_d|/|A_{cm}|)$

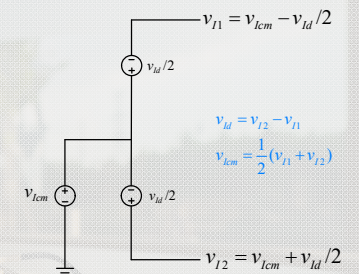


Fig. Representing the input signals to a differential amplifier in terms of their differential and common-mode components.

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Differential Amplifier: Differential gain



■ Apply superposition method

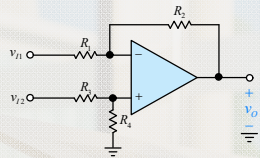


Fig. A difference amplifier.

$$v_{O1} = -\frac{R_2}{R_1} v_{I1} \quad \text{Condition: } \frac{R_4}{R_3} = \frac{R_2}{R_1}$$

$$v_{O2} = \left(\frac{R_4}{R_3 + R_4} \right) \left(1 + \frac{R_2}{R_1} \right) v_{I2} = \frac{R_2}{R_1} v_{I2}$$

■ Differential gain

$$v_O = \frac{R_2}{R_1} (v_{I2} - v_{I1}) = \frac{R_2}{R_1} v_{Id}$$

$$A_d = \frac{R_2}{R_1}$$

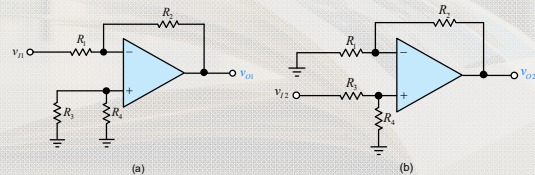
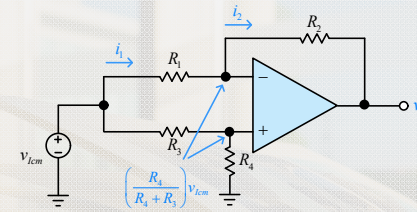


Fig. Superposition analysis

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Differential Amplifier: common mode gain



$$A_{cm} \equiv \frac{v_O}{v_{Icm}} = \left(\frac{R_4}{R_3 + R_4} \right) \left(1 - \frac{R_2 R_3}{R_1 R_4} \right)$$

$$\text{Condition: } \frac{R_4}{R_3} = \frac{R_2}{R_1}$$

$$A_{cm} = 0$$

Fig. Analysis of the difference amplifier to determine its common-mode gain $A_{cm} \approx v_O/v_{cm}$

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Differential Amplifier: input resistance

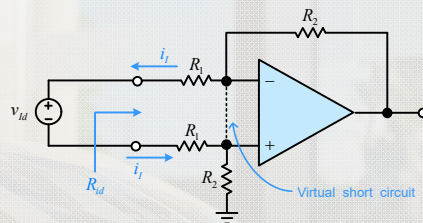


Fig. Finding the input resistance of the difference amplifier for the case $R_3 = R_1$ and $R_4 = R_2$.

$$R_{id} = 2R_1$$

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